

1 Bilinear finite-part Stieltjes correlations

Let $\zeta(1+u, x) = u^{-1} + \sum_{j \geq 0} (-1)^j \gamma_j(x) u^j / j!$, so $\gamma_0 = -\psi$. For $0 < a < 1$ put $b = 1 - a$ and define

$$I_{mn}(a) = \lim_{\epsilon \downarrow 0} \left\{ \int_{\epsilon}^b \gamma_m(x) \gamma_n(a+x) dx + \int_{\epsilon}^a \gamma_m(b+y) \gamma_n(y) dy + \frac{\gamma_n(a)}{m+1} \log^{m+1} \epsilon + \frac{\gamma_m(b)}{n+1} \log^{n+1} \epsilon \right\}. \quad (1)$$

This is the finite part of a circular translation, not an ordinary integral. The identity $\gamma_m(x) = x^{-1} \log^m x + \gamma_m(1+x)$ proves that the limit exists: subtract the two displayed local singular terms inside the integrals, then add their integrated upper-endpoint terms. The resulting integrals are absolutely convergent.

Theorem 1.1 (All-order bilinear closure). *For $m, n \geq 0$ and $N = m + n + 1$,*

$$I_{mn}(a) = \frac{\gamma_N(a)}{m+1} + \frac{\gamma_N(b)}{n+1} + e_{mn} + \sum_{j=0}^{N-2} (c_{mn,j} \gamma_j(a) + d_{mn,j} \gamma_j(b)), \quad (2)$$

where the sum is empty for $N = 1$. All coefficients are explicit polynomials over $\mathbb{Q}$ in $\zeta(2), \dots, \zeta(N+1)$. There is no γ_{N-1} term. Also $I_{mn}(a) = I_{nm}(1-a)$.

Proof. Put $w = u + v$ and

$$\mathcal{A}_{cc}(u, v) = \frac{\Gamma(1+w)\Gamma(1-v)}{\Gamma(1+u)}, \quad \mathcal{B}_{cc}(u, v) = \mathcal{A}_{cc}(v, u), \quad \mathcal{G}(u, x) = \zeta(1+u, x) - \frac{1}{u}.$$

Classical Hurwitz Fourier orthogonality, initially in an absolutely convergent spectral region, and Gamma reflection give the kernel

$$\mathcal{K} = -u \mathcal{A}_{cc}(u, v) \zeta(1+w, b) - v \mathcal{B}_{cc}(u, v) \zeta(1+w, a). \quad (3)$$

It is uv times the meromorphic continuation of the circular product integral of $\zeta(1+u, x)$ and $\zeta(1+v, x \oplus a)$. For an explicit endpoint justification, that product's regular part is

$$R = \int_0^b \{x^{-1-u} [\mathcal{G}(v, a+x) - \mathcal{G}(v, a)] + \mathcal{G}(u, 1+x) \mathcal{G}(v, a+x)\} dx + \int_0^a \{y^{-1-v} [\mathcal{G}(u, b+y) - \mathcal{G}(u, b)] + \mathcal{G}(v, 1+y) \mathcal{G}(u, b+y)\} dy + \mathcal{G}(v, a) \frac{1-b^{-u}}{u} + \mathcal{G}(u, b) \frac{1-a^{-v}}{v}.$$

The quotients take their removable values $\log b$ and $\log a$. The differences vanish linearly at the respective endpoints, so R is holomorphic near zero, including after spectral differentiation. Its coefficients are exactly $(-1)^{m+n} I_{mn} / (m!n!)$. Using zero mean of the Hurwitz function in the initial convergence region identifies R as the mixed divided difference of $\mathcal{K}$. Hence

$$I_{mn} = (-1)^{m+n} m!n! [u^{m+1} v^{n+1}] \mathcal{K}.$$

Since $\mathcal{A}_{cc}(u, -u) = \mathcal{B}_{cc}(u, -u) = 1$, the quotient $Q = (u\mathcal{A}_{cc} + v\mathcal{B}_{cc})/(u + v)$ is holomorphic, and

$$\mathcal{K} = -Q - u\mathcal{A}_{cc}\mathcal{G}(w, b) - v\mathcal{B}_{cc}\mathcal{G}(w, a).$$

Finally,

$$\log \mathcal{A}_{cc} = \sum_{k \geq 2} \frac{\zeta(k)}{k} \{(-1)^k ((u + v)^k - u^k) + v^k\}.$$

This proves finite linear closure and the coefficient algebra. The constant Gamma-ratio term gives the two coefficients of γ_N ; the missing degree-one Gamma-ratio term gives the absence of γ_{N-1} . Interchanging the variables proves symmetry. The accompanying report supplies the full Fourier derivation and the exact finite coefficient algorithm. $\square$

Corollary 1.2 (First identities and an antiderivative). *For $0 < a < 1$,*

$$\begin{aligned} I_{00}(a) &= \gamma_1(a) + \gamma_1(1 - a) - \frac{\pi^2}{3}, \\ I_{10}(a) &= \frac{1}{2}\gamma_2(a) + \gamma_2(1 - a) + \zeta(2)(\gamma_0(a) + \gamma_0(1 - a)) - \zeta(3), \\ \int I_{00}(a) da &= \frac{1}{2}(\zeta''(0, a) - \zeta''(0, 1 - a)) - \frac{\pi^2}{3}a + \text{constant}. \end{aligned}$$

Proof. Expand the kernel through total degrees two and three. For the primitive, use $\partial_a \zeta^{(j+1)}(0, a) = (-1)^{j+1}(j + 1)\gamma_j(a)$. $\square$

Remark 1.3 (Evidence and normalization). The accompanying delivery proves the coincident-endpoint analogue and its collision counterterm, an ordinary log-Gamma autocorrelation, arithmetic traces, and higher derivative and primitive formulas. Its exact table covers $m + n \leq 8$, but the theorem above holds at all orders by proof. Numerical checks are diagnostics, not interval certificates, and no period independence or literature priority is asserted. The classical starting identities are documented in DLMF Section 25.11.